

Debugging Assignment – Bug Fix Report

Objective

Identify and fix bugs in the provided Express.js application to ensure all APIs function correctly and the server runs without errors.

Bug 1: Undefined Variable in `/users`

Original Code

```
const allUsers = users;  
res.send(userList);
```

Issue

`userList` is not defined, causing a runtime error.

Fix

```
res.send(users);
```

Bug 2: Type Mismatch in `/users/:id`

Original Code

```
const id = req.params.id;  
const user = users.find(u => u.id === id);
```

Issue

`req.params.id` is a string while `u.id` is a number.

Fix

```
const id = Number(req.params.id);  
const user = users.find(u => u.id === id);
```

Bug 3: Missing Return Statement

Original Code

```
function getUserById(id) {  
  const user = users.find(u => u.id === id);  
}
```

Issue

Function does not return the found user.

Fix

```
function getUserById(id) {  
  return users.find(u => u.id === id);  
}
```

Bug 4: Typographical Error in **/notes/count**

Original Code

```
const total = notes.lenght;
```

Issue

Incorrect property name (**lenght**).

Fix

```
const total = notes.length;
```

Bug 5: Undefined Async Function in `/external-data`

Original Code

```
const data = fetchExternalData();
```

Issues

1. Function is not defined.
2. Promise is not awaited.

Fix

```
const data = await fetchExternalData();
```

Bug 6: Assignment Instead of Comparison

Original Code

```
if (notes = []) {
```

Issue

Assignment operator used instead of checking array length.

Fix

```
if (notes.length === 0) {
```

Bug 7: Non-Integer Note ID

Original Code

```
return Math.random() * 1000;
```

Issue

Returns decimal values.

Fix

```
return Math.floor(Math.random() * 1000);
```

Bug 8: Function Reference Stored Instead of Value

Original Code

```
const newId = generateNotId;
```

Issue

Stores function reference rather than generated ID.

Fix

```
id: generateNotId()
```

Bug 9: Incorrect Input Validation

Original Code

```
if (!title && !content)
```

Issue

Validation fails only when both fields are missing.

Fix

```
if (!title || !content)
```

Bug 10: Invalid Note ID Assignment

Original Code

```
id: newId
```

Issue

Assigns function instead of generated value.

Fix

```
id: generateNotId()
```

Bug 11: Type Mismatch in Delete API

Original Code

```
const noteIndex = notes.findIndex(n => n.id === id);
```

Issue

`id` is a string.

Fix

```
const id = Number(req.params.id);  
const noteIndex = notes.findIndex(n => n.id === id);
```

Bug 12: Undefined Variable in Update User API

Original Code

```
user.name = username;
```

Issue

`username` is not defined.

Fix

```
user.name = name;
```

Bug 13: Assignment Instead of Comparison in Filter

Original Code

```
notes.filter(n => n.userId = userId);
```

Issue

Uses assignment operator.

Fix

```
notes.filter(n => n.userId === userId);
```

Bug 14: Authentication Logic Error

Original Code

```
if (email === "admin@test.com" || password === "123456")
```

Issue

Login succeeds if either email or password matches.

Fix

```
if (email === "admin@test.com" && password === "123456")
```

Bug 15: Incorrect Use of `filter()` in Profile API

Original Code

```
const user = users.filter(u => u.id === id);  
res.send(user.name);
```

Issue

`filter()` returns an array, which does not have a `name` property.

Fix

```
const user = users.find(u => u.id === id);  
res.send(user);
```

Bug 16: String Concatenation in Sum API

Original Code

```
const total = a + b;
```

Issue

If inputs are strings, concatenation occurs.

Fix

```
const total = Number(a) + Number(b);
```

Bug 17: Incorrect Server Port Message

Original Code

```
app.listen(3000, () => {  
  console.log("Server running on port 5000");  
});
```

Issue

Log message does not match actual listening port.

Fix

```
app.listen(3000, () => {  
  console.log("Server running on port 3000");  
});
```